Say It as a Premise, Not a Shortlist: Which Form of a Team-Wide Note Helps Wrong LLM Nodes Without Hurting Right Ones

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Abstract

A note broadcast to a whole team of language-model "nodes" cannot be targeted: it reaches nodes that already answered correctly and nodes that answered wrongly, and the sender knows neither which is which nor whether the note itself is true. We ask which *form* of such a blind broadcast comes closest to helping the wrong nodes without hurting the right ones. For the same question we write a true and a false note in each form, give each to every node after it has answered once, and measure the keep rate K of nodes that were right and the rescue rate B of nodes that were wrong; the two combine into v^*, the smallest probability that the note is true at which broadcasting it has non-negative expected value. Across five pre-registered rounds on six non-Claude nodes, an 80-item and a 297-item set of multiple-choice questions, and 80 open-ended rewrites, we find: (i) a false *shortlist* of candidate answers is far more destructive than a false *premise* about the same item (K 29% vs. 86%); (ii) appending "unless you have strong reasons for another option" to the shortlist recovers most of the lost keep in every setting at a small rescue cost, but does nothing for premises; (iii) by v^* the premise is cheaper than either shortlist form, a margin that holds on every node once the echo of the node's earlier answer is removed; (iv) the premise is not cheaper than simply asserting one answer: it keeps far more right nodes ($+.58$) and rescues far fewer wrong ones ($-.63$), with no detectable difference in v^*; (v) with the source passage available, nodes that abandon a correct answer under a false shortlist justify the switch with a genuine quotation 94% of the time: they pick first and then find support. We release items, notes, raw responses and the pre-registrations.

1 Introduction

Multi-agent systems built from large language models (LLMs) increasingly share information by broadcast: a coordinator posts a hint, a memory, a retrieved fact or a shortlist of candidate answers to every worker (Du et al., 2024). The broadcast is *blind* in two senses. The sender does not know which workers are currently right, and the sender does not know for certain whether the note itself is true. A useful note must therefore satisfy two requirements at once: when it is true it should move the workers that were wrong, and when it is false it should not move the workers that were right.

These two requirements are usually studied separately. Work on sycophancy and misleading context measures how easily an LLM abandons a correct answer under pressure (Sharma et al., 2024; Wang et al., 2023; Xu et al., 2024a); work on hints and advice measures how much a wrong model benefits from help (Bailey et al., 2023). Neither literature asks the question a broadcaster faces: given that I must send the *same sentence* to everyone, and that I may be wrong, which form of sentence should I send?

We answer this with a paired design. For each item we write a *true* and a *false* version of the note in every form under test; the false version points at a specific wrong option. Every node answers each item once without a note, then again with each note. On the items a node got right we measure the **keep rate** K (fraction that stay right); on the items it got wrong we measure the **rescue rate** B (fraction that become right). Combining the true-note and false-note effects with the node's own error rate yields v^*, the break-even truth probability of the note: a form with lower v^* is safer to broadcast.

We compare four forms that differ in what they assert. A *shortlist* names two candidate options ("the correct answer is one of: B, D."). A *premise* states one key fact or step from which the answer follows, without naming an option. Each may carry an *exit clause*: "..., unless you have strong reasons for another option" for shortlists, "Rely on

this unless you have strong reasons to think otherwise." for premises. A control round adds a bare *assertion* of one option ("the correct answer is B.") as a strength baseline.

Our contributions are:

- A same-item paired true/false design and the break-even measure v^* that turn "helps the wrong without hurting the right" into a single pre-registrable number (§2).
- Evidence that a false shortlist is the most destructive form (keep 29% on the 80-item set, 18% on its hard half), that a false premise about the same item costs far less (86%), and that the gap is specific to the shortlist form rather than to multiple-choice formatting (§4).
- A pre-registered test, replicated on the 80-item set, its 80 open-ended rewrites and 297 fresh items, showing that an exit clause on a shortlist recovers most of the lost keep at a small rescue cost, while the same clause on a premise gives no gain (§5).
- A head-to-head by v^* (§6) and two pre-registered controls (§7): the premise is cheaper than both shortlist forms, with a margin over shortlist-with-exit that depends on a definitional choice when the node's earlier answer is echoed and holds on every node when it is not; against a bare single-option assertion the premise is far safer when false and far weaker when true, with no difference in v^*.
- A mechanism probe on reading comprehension: giving nodes the passage halves the damage of a false shortlist, yet nodes that still switch away quote a true sentence to support the wrong option; the shortlist turns "read and answer" into "pick and justify" (§8).

All rounds were pre-registered in a public repository before the data were collected; criteria, thresholds and the decision table were fixed in the commit that launched each run (repository URL withheld for review). Analyses that were not pre-registered are labelled exploratory.

2 Measures

Let a node answer an item once (the *first answer*) and then again after seeing a note p (the *second answer*). For node a and note form p,

$$K_a(p) = \Pr(\text{second right} \mid \text{first right}),$$
$$B_a(p) = \Pr(\text{second right} \mid \text{first wrong}).$$

The blank condition ($p = \varnothing$) gives the baselines K_a^0, B_a^0; in all our rounds $K^0 \geq 0.98$ for multiple-choice items, so a node repeating itself without a note keeps its answer.

Every form p comes in a true version p_t and a false version p_f for the same item. Let ε_a be the node's first-answer error rate. The expected change in accuracy from broadcasting the true version is

$$n_t = \varepsilon_a \left[B_a(p_t) - B_a^0\right] - (1 - \varepsilon_a)\left[K_a^0 - K_a(p_t)\right],$$

and n_f is defined likewise with p_f. If the sender believes the note is true with probability v, the expected gain is $v\, n_t + (1 - v)\, n_f$, which is non-negative iff

$$v \;\geq\; v^* = \frac{-n_f}{n_t - n_f}.$$

v^* is the break-even truth probability: a form with $v^* = 0.84$ should only be broadcast when the sender is at least 84% sure it is right, whereas $v^* = 0.4$ can be sent on a weaker belief. We clip v^* to 0 when the false version does no net harm ($n_f \geq 0$) and to 1 when the true version does no net good ($n_t \leq 0$); the unclipped version is reported alongside and is undefined in those cases. All statistics are node means; confidence intervals are 95% percentile intervals from 2,000 bootstrap resamples over items, with paired differences computed within node and item.

3 Setup

Items. Multiple-choice items come from MMLU (Hendrycks et al., 2021) subsets grouped into three task types: *recall* (global facts, prehistory, astronomy, miscellaneous, US foreign policy), *reason* (high-school and college mathematics, statistics, physics, formal logic, abstract algebra, econometrics) and *case* (professional and international law, jurisprudence, clinical knowledge, college and professional medicine). We exclude items whose options refer to other options ("all of the above", "both A and B") and, for the open-ended rewrites, items that cannot be answered without seeing the options.

Reading-comprehension items come from RACE (Lai et al., 2017). Three item sets are used:

- **80-item set**: 40 *hard* items on which at least one frontier model erred in an earlier screening, and 40 *fresh* items from the same sources with no screening (case 56, recall 24).
- **297-item set**: from a pool of 1,366 unused items, five inexpensive nodes screened for items that at least one node got wrong (635 candidates); 320 were drawn balanced across the three task types, and 297 survived note generation (case 103, reason 98, recall 96).
- **RACE set**: 51 passages with one four-option question each.

Nodes. We treat each model configuration as a *node*. The main rounds use six nodes: DeepSeek V4 Flash at temperature 0 with a "precise assistant" system prompt (FLASH-A), DeepSeek V4 Flash at temperature 1 with an "answer quickly from memory" prompt (FLASH-B; the only sampled node), DeepSeek V4 Pro at temperature 0 with reasoning disabled (PRO), OpenAI Codex CLI with web search disabled (CODEX), and, in the earlier rounds, Kimi K3.1 Pro and Grok 4.7 through their command-line interfaces with tools and web access disabled. The 297-item round replaces Kimi and Grok, whose quotas were exhausted mid-run (7% and 12% coverage, excluded by the pre-registered coverage check), with two open-weight models served locally at temperature 0: Qwen2.5-14B-Instruct (QWEN) and Llama-3.1-8B-Instruct (LLAMA). Claude Sonnet and Haiku appear only in the format study and the exploratory open-ended round. No node has conversational memory between calls.

Protocol. Items are presented in batches of three to five. The first prompt asks for a JSON object per item with an answer letter, a confidence and a one-sentence reason. The second prompt repeats the items, echoes the node's own earlier answer, confidence and reason, and appends "A note shared with the whole team: …". The note is identical for every node. The control round (§7) also runs a variant of the second prompt without the echo, so that the node sees only the items and the note. For open-ended rewrites the options are removed, the node answers in free text, and a separate DeepSeek V4 Pro judge, blind to node and note, grades the stated conclusion against the reference (Cohen's $\kappa = .83$ against a Codex re-grade of 100 rows, .88 against an earlier Claude Sonnet grade of 1,600 rows). Full prompts are in Appendix A.

Form	Note text (false version shown where it differs only in the options named)
ONE1	the correct answer is B.
SET2	the correct answer is one of: B, D.
SOFT2	the correct answer is probably one of: B, D.
COND2	the correct answer is one of: B, D, unless you have strong reasons for another option.
SET3	the correct answer is one of: A, B, D.
L3	*[one-sentence premise]*
L3C	*[premise]* Rely on this unless you have strong reasons to think otherwise.

Table 1: Note forms. Each has a true (t) and false (f) version for the same item. ONE1 false names the designated wrong target. SET3 true contains the gold option and two wrong ones; false contains three wrong ones.

Forms. Table 1 lists the seven forms. Six are used in the main rounds; with the true and false versions and the blank control they give nine conditions. The seventh, ONE1, names a single option and is used only in the control round (§7). The false shortlist contains two wrong options; the true shortlist contains the gold option and one wrong option. Premises were written by DeepSeek V4 Pro from the item, its gold option and a designated wrong target: the true premise is a key step from which a careful reader can derive the answer, the false premise is a plausible step that leads to the target. A program rejects premises that name an option letter, quote the gold (true) or target (false) option text, or are shorter than 20 characters; 23 of 320 items were dropped on this check in the 297-item round. The same premise sentences are used across all nodes and all rounds that share an item.

Pre-registration. Each round's script, item list, thresholds and decision table were committed to a public repository before the run; the results commit follows. Criteria are of the form "node-mean difference $\geq \theta$, bootstrap lower bound > 0, and at least k of n nodes in the same direction". We did not correct for multiple comparisons and report every pre-registered test; exploratory analyses are marked as such.

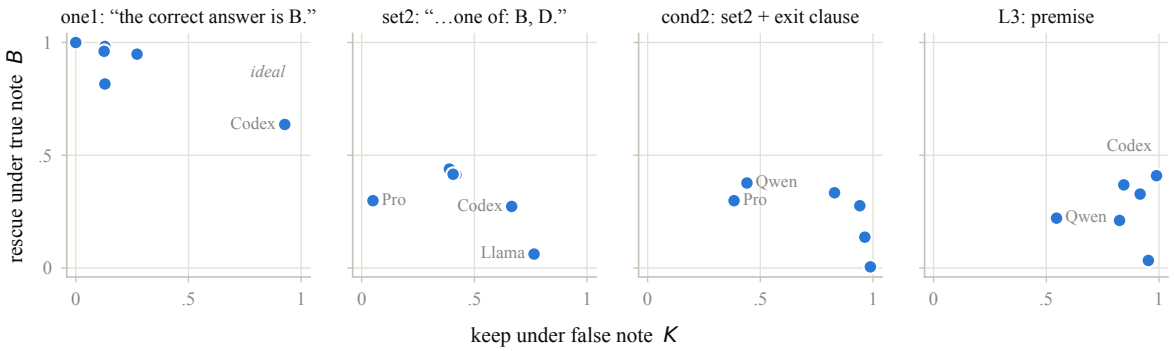


Figure 1: The trade each form makes: keep under the false note against rescue under the true note, one point per node, 297-item round with echo (ONE1 from the control round, §7). A bare assertion moves almost every node, right or wrong; a shortlist loses most right nodes for a modest rescue; the exit clause restores keep on every node but QWEN; a premise keeps right nodes on all but one node, at a cost in rescue.

4 A false shortlist destroys correct answers; a false premise mostly does not

Table 2 gives node-mean keep and rescue rates for every form in the three multiple-choice rounds. Two facts stand out. First, the *true* versions of every form rescue a substantial share of wrong answers (true shortlist 57% on the 80 hard items, true premise 55%), and all keep correct answers almost perfectly. Second, the *false* versions differ sharply: a false shortlist keeps only 29% of correct answers on the 80-item set (18% on its 40 hard items) and 45% on the fresh 297, whereas a false premise about the same item keeps 86% (90% on the hard half) and 85%.

Is this a multiple-choice artefact? A pre-registered format study on the 80-item set with seven nodes (Kimi, Grok, FLASH-A, PRO, CODEX, Claude Sonnet and Claude Haiku) presented each item in three conditions: the original batched multiple-choice format (J), single items with options and a three-sentence free-text answer (M), and the same prompt with the options removed (O). In M and O the false shortlist named the option *texts*. Removing the options reduced the damage of the false shortlist only insignificantly (+.09 [−.02, +.20], 3/7 nodes), and the batched-JSON format added only +.08 [−.02, +.17]. The pre-registered *form-specificity* test, which asks whether the J−O difference is larger for the shortlist than for the premise, passed (+.24 [+.11, +.37]): the shortlist loses .21 more keep in the batched format while the premise loses nothing (−.02). We therefore treat the destructiveness of a false shortlist as a property of the form, modulated but not created by the answer format.

Exploratory: on the 80-item set every right node that abandoned its answer under a false shortlist moved *into* the shortlist, never to the unlisted wrong option.

5 An exit clause rescues the shortlist, not the premise

5.1 Wording round (80 items, 6 nodes)

The wording round pre-registered three criteria for the exit clause: W1 keep gain $K(\text{COND2}_f) - K(\text{SET2}_f) \geq .15$ with lower bound > 0 and $\geq 4/6$ nodes positive; W2 rescue cost $B(\text{COND2}_t) - B(\text{SET2}_t) \geq -.15$ with upper bound < 0 not required; W3 $v^*(\text{COND2}) - v^*(\text{SET2}) \leq -.10$ with upper bound < 0. All three passed (Table 3): the clause recovers about two-thirds of the keep lost to a false shortlist (+.50 of .71, every node, range +.21 to +.83), costs .14 rescue, and lowers the break-even truth probability from .84 to .60. The rescue cost sits on the threshold: Grok and PRO lost no rescue, and in the pre-registered sensitivity check that drops the two CLI nodes (Grok, CODEX) the cost is −.19.

Two alternatives were run in the same round. Hedging ("probably one of", SOFT2) behaves like the exit clause (+.44 keep, −.11 rescue, v^* .84 → .64). Widening the shortlist to three options (SET3) is worse than the exit on both sides: it recovers only +.18 keep and loses .35 rescue, raising v^* to .91. Giving the right answer more room to survive is not the same as giving the node permission to leave.

Form	80 items (40 hard) (wording round) K	B	80 items (40 hard) (final-form round) K	B	297 fresh items (scale round) K	B
blank	100	0	100	0	100	5
SET2$_t$	99	57	99	57	97	32
SET2$_f$	**29**	0	**29**	0	**45**	3
SOFT2$_t$	99	46	–	–	–	–
SOFT2$_f$	72	0	–	–	–	–
COND2$_t$	100	42	100	42	98	24
COND2$_f$	**79**	0	**79**	0	**76**	5
SET3$_t$	99	21	–	–	–	–
SET3$_f$	47	0	–	–	–	–
L3$_t$	–	–	100	55	99	26
L3$_f$	–	–	**86**	5	**85**	7
L3C$_t$	–	–	99	59	100	32
L3C$_f$	–	–	83	2	81	7

Table 2: Node-mean keep rate K (% of first-right answers kept) and rescue rate B (% of first-wrong answers corrected) by form. The two 80-item rounds share items, first answers and nodes (Kimi, Grok, FLASH-A, FLASH-B, PRO, CODEX); the scale round replaces Kimi and Grok with QWEN and LLAMA.

Round	Criterion	Estimate [95% CI]	Nodes	Result
80, MC	keep gain $K(\text{COND2}_f) - K(\text{SET2}_f)$	$+.50$ [$+.46$, $+.55$]	6/6 > 0	pass
	rescue cost $B(\text{COND2}_t) - B(\text{SET2}_t)$	$-.14$ [$-.23$, $-.05$]	6	pass (threshold $-.15$)
	$v^*(\text{COND2}) - v^*(\text{SET2})$	$-.24$ [$-.33$, $-.16$]	6	pass
80, open-ended	keep gain	$+.23$ [$+.18$, $+.29$]	6/6 > 0	pass
	rescue cost	$-.03$ [$-.06$, $-.00$]	6	pass
	v^* difference	$-.16$ [$-.20$, $-.11$]	6	pass
297 fresh, MC	keep gain	$+.31$ [$+.28$, $+.34$]	6/6 > 0	pass
	rescue cost	$-.08$ [$-.11$, $-.05$]	6	pass
	v^* difference	$-.12$ [$-.25$, $-.01$]	6	pass
80, MC	exit on premise, keep $K(\text{L3C}_f) - K(\text{L3}_f)$	$-.03$ [$-.06$, $+.00$]	2/6 > 0	fail
	exit on premise, rescue $B(\text{L3C}_t) - B(\text{L3}_t)$	$+.05$ [$+.00$, $+.09$]	6	no net gain
297 fresh, MC	exit on premise, keep	$-.03$ [$-.05$, $-.01$]	2/6 > 0	fail
	exit on premise, rescue	$+.06$ [$+.03$, $+.09$]	6	no net gain

Table 3: Pre-registered tests of the exit clause. Keep gain is measured on false notes, rescue cost on true notes; all differences are paired within node and item. "Nodes" gives how many of the available nodes had the difference in the reported direction (keep) or the number of nodes with at least 8 (15 in the scale round) first-wrong items (rescue).

5.2 Replications

The exit result replicated twice with the criteria fixed in advance. On the 80 items rewritten as open-ended questions (no options; the note names option texts), the clause gains $+.23$ keep on every node at a rescue cost of .03, and lowers v^* by .16. On the 297 fresh items with two open-weight nodes replacing Kimi and Grok, the gain is $+.31$ on every node, the cost .08, and the v^* reduction .12. The gain is smaller on the small open-weight models (QWEN $+.03$, LLAMA $+.22$) than on the API nodes ($+.30$ to $+.52$).

5.3 The clause does nothing for premises

The final-form round added the exit clause to premises (L3C). On both the 80-item and the 297-item sets the clause slightly *increases* the damage of a false premise ($-.03$ keep, lower bound below zero in the larger set) and slightly increases the rescue of a true premise ($+.05$, $+.06$). The two cancel: $v^*(\text{L3C}) - v^*(\text{L3})$ is $-.03$ [$-.09$, $+.04$] and $-.03$ [$-.11$, $+.03$]. The exit clause works by licensing the node to step outside a shortlist; a premise has no inside to step out of.

6 Premise versus shortlist-with-exit

Two definitions of v^*. The ratio $-n_f/(n_t - n_f)$ is undefined when a false note does a node no net harm ($n_f \geq 0$) or a true note does it no net good ($n_t \leq 0$). In the 80-item round this happened for Grok and CODEX under the false premise: it did not hurt them, so their $v^*(\text{L3})$ could not be com-

Round	SET2	COND2	L3	L3C
clipped definition				
80 items, 6 nodes	.84	.60	.38	.42
297 fresh, 6 nodes	.84	.71	.54	.51
API nodes only (4)	.92	.78	.55	.59
open-weight only (2)	.66	.58	.52	.37
case items	.85	.75	.47	.46
reason items	.80	.57	.39	.50
recall items	.86	.64	.58	.55
no echo (§7)	.75	.68	.50	–
unclipped definition (defined nodes only)				
80 items	.84	.60	.57 (4)	.54 (4)
297 fresh	.84	.71	.65 (5)	.51 (6)
no echo	.75	.68	.50	–

Table 4: Break-even truth probability v^* (node means). Lower is cheaper to broadcast. Under the unclipped definition the number of nodes with a defined value is given in parentheses when it is fewer than six. The clipped 80-item row was computed after the fact with the definition pre-registered for the 297-item round. The no-echo rows come from the control round, which did not run L3C; the bare assertion ONE1 (with echo) scores .62 under both definitions, all six nodes defined.

puted and the pre-registered comparison ran on four nodes. For the 297-item round we therefore pre-registered the clipped definition of §2 ($v^* = 0$ when $n_f \geq 0$, 1 when $n_t \leq 0$), which assigns a node that a false note cannot harm the lowest possible break-even probability, and committed to report the unclipped version alongside. The change was made after, and because of, the undefined cases in the first round; we report both throughout.

Results. Table 4 gives v^* for the four main forms. Under either definition and in every round, the plain shortlist is the most expensive of the four and the shortlist with exit is next; the premise, with or without exit, is cheaper on average, and the two premise forms are not distinguishable from each other (their difference is $+.04$ $[-.03, +.11]$ on the 80-item set and $-.03$ $[-.11, +.03]$ on the 297-item set, clipped). The pre-registered head-to-head comparisons are:

- **80 items, unclipped (pre-registered).** $v^*(\text{L3C}) - v^*(\text{COND2}) = -.18$ $[-.28, -.02]$, 6 nodes: pass. $v^*(\text{L3}) - v^*(\text{COND2}) = -.11$ $[-.24, +.03]$ on the four defined nodes: fail. Clipped, after the fact: $-.22$ $[-.34, -.09]$ and $-.18$ $[-.31, -.05]$, 6 nodes.
- **297 fresh, clipped (pre-registered).** $v^*(\text{L3}) - v^*(\text{COND2}) = -.17$ $[-.29, -.01]$, 6 nodes: pass. Unclipped: $-.06$ $[-.12, +.02]$ on the five defined nodes (CODEX undefined): would fail. $v^*(\text{L3C}) - v^*(\text{COND2}) = -.20$ $[-.32, -.06]$, 6 nodes, under both definitions.

The pass therefore hinges on how a node that a false premise cannot harm is counted. On the 297-item set the false premise left CODEX 99% of its correct answers and did it no net harm; the clipped definition scores that as $v^* = 0$ against .75 for the shortlist-with-exit, and that single node carries the mean. On the remaining nodes the two forms are close: LLAMA (.40 vs. .50) and PRO (.78 vs. .90) favour the premise modestly, FLASH-B (.78 vs. .84) slightly, and FLASH-A (.65 vs. .64) and QWEN (.65 vs. .65) are tied. By task type the premise advantage is clear on case items ($-.28$ $[-.40, -.12]$) and not resolved on reasoning or recall items.

Neither component is decisive on its own: relative to shortlist-with-exit, the false premise is kept slightly better ($+.08$ and $+.09$ keep, 4/6 nodes positive) and the true premise rescues about as well ($+.12$ $[-.09, +.33]$ and $+.02$ $[-.03, +.08]$). We state the finding at the level the data support: no node is clearly cheaper to serve with a shortlist-with-exit than with a premise; three nodes are tied or nearly so; and the premise is clearly cheaper on the nodes that a false premise does not move at all. Whether that last group is the rule or the exception among models is the open question. Section 7 revisits both points.

7 Two pre-registered controls

A fifth pre-registered round on the 297-item set and the same six nodes tested two weaknesses of the rounds above: form is confounded with strength (a false shortlist excludes two options with certainty; the persuasiveness of a model-written premise is uncalibrated), and every second answer echoed the node's own first answer, so every effect was measured against a self-anchored prior. The round added a seventh form, ONE1 ("the correct answer is B."), and re-ran BLANK, SET2, COND2 and L3 with the echo removed from the second prompt. Pre-registered criteria: E1b, the premise is *safer* than the bare assertion (keep difference $\geq .15$, lower bound > 0, at least four nodes positive); E1c, it is *cheaper* (upper bound of the v^* difference below zero); E2b and E2c, the exit and premise keep gains pass the same keep criterion without the echo. Passing E1b but fail-

ing E1c was to be read as "safer when false, not cheaper".

A bare assertion is stronger in both directions. True, the single-option assertion rescues 64–100% of wrong nodes ($B(\text{ONE1}_t) - B(\text{SET2}_t) = +.57$ [$+.52$, $+.62$]); false, it destroys 73–100% of correct answers on five nodes ($K(\text{ONE1}_f)$ 0–27%), CODEX being the exception at 93%. The two-option shortlist is a weaker version of the same move: it does somewhat less damage when false ($K(\text{ONE1}_f) - K(\text{SET2}_f) = -.18$ [$-.21$, $-.16$]) but rescues far less when true, and by v^* it is the worse deal ($v^*(\text{SET2}) - v^*(\text{ONE1}) = +.21$ [$+.18$, $+.25$], clipped). Shortlist-with-exit is not distinguishable from the bare assertion ($+.09$ [$-.03$, $+.21$]).

The premise is safer than a bare assertion, not cheaper. E1b passes: $K(\text{L3}_f) - K(\text{ONE1}_f) = +.58$ [$+.55$, $+.61$], all six nodes positive (CODEX only $+.06$, since it ignores both). But the true premise rescues far less than the true assertion, $B(\text{L3}_t) - B(\text{ONE1}_t) = -.63$ [$-.67$, $-.59$], and the two effects cancel in the break-even probability: $v^*(\text{L3}) - v^*(\text{ONE1}) = -.08$ [$-.14$, $+.01$] clipped on six nodes, $+.02$ [$-.05$, $+.09$] unclipped on five. E1c fails, and with it the pre-registered conclusion "the strength confound cannot explain the premise advantage". The premise's v^* advantage in §6 is an advantage over *shortlists*, which rescue poorly for the damage they do, not over naming the answer outright. What the premise form buys is a different trade, not a cheaper one: it moves far fewer nodes, right or wrong, which is also what a weaker push would produce.

Without the echo, the ordering sharpens. Removing the echo changes the baseline: blank keep falls to 82–100% (LLAMA 82%) and blank rescue rises to 4–26% (FLASH-B 26%). Both keep tests pass on all six nodes: $K(\text{COND2}_f) - K(\text{SET2}_f) = +.15$ [$+.13$, $+.18$] (E2b, on the pre-registered threshold) and $K(\text{L3}_f) - K(\text{SET2}_f) = +.19$ [$+.15$, $+.23$] (E2c), with a rescue cost of $-.05$ [$-.09$, $-.01$] for the exit clause and a rescue gain of $+.06$ [$+.00$, $+.11$] for the premise. The keep margin of the premise over the plain shortlist is thus about half of what it is with the echo ($+.40$ in Table 2). The remaining quantities were pre-registered as reports, not decision criteria. Every v^* is lower without the echo (Table 4: SET2 .75, COND2 .68, L3 .50), and the premise's margin over shortlist-with-exit, which in §6 depended on the clipped definition, now holds under both: $-.18$ [$-.24$, $-.11$] clipped and $-.18$ [$-.23$, $-.07$] unclipped, with all six nodes defined (CODEX .23 vs. .80). The echo itself pushes the two forms in opposite directions: with it, the four API nodes keep 19–33 points *less* under a false shortlist ($-.09$ [$-.12$, $-.05$] over six nodes, LLAMA $+.65$) and 10–13 points *more* under a false premise ($+.12$ [$+.09$, $+.15$], 5/6 nodes), and rescue is lower with the echo on every form ($-.11$ to $-.22$): a node reminded of its wrong answer is harder to move.

8 Mechanism: pick, then justify

Why does a false shortlist override a correct answer that a premise leaves intact? One hypothesis is that nodes abandon answers they cannot verify, so giving them the evidence should protect them. We tested this on RACE reading comprehension, where the evidence is a passage. Six nodes answered 51 questions with the passage, then answered again with a note under two conditions that differ in one line: the passage is shown (E) or replaced by "The passage is not shown this time." (R). The reply format asked for the single sentence of the passage the node relied on most, quoted verbatim or left empty.

Evidence helps. The pre-registered criterion Q1 asks whether the damage of a false shortlist is larger without the passage: $[K_R^0 - K_R(\text{SET2}_f)] - [K_E^0 - K_E(\text{SET2}_f)] = +.39$ [$+.31$, $+.47$], all six nodes positive (Kimi only $+.10$: it keeps 10% even with the passage). The false premise shows the same pattern more strongly ($+.52$ [$+.45$, $+.59$]): without the passage a false claim about its content is believed.

But the nodes that are fooled are not fabricating. Criterion Q2 asked whether, with the passage shown, rows that abandon the correct answer under a false shortlist are accompanied by empty or fabricated quotations more often than rows that keep it. They are not: 94% of the 140 abandoning rows quote a genuine sentence of the passage, against 97% of the keeping rows ($+.03$ [$-.02$, $+.08$]). An exploratory look at the abandoning rows shows that 81% did not re-quote the sentence they had relied on at first, most quoting a different genuine sentence, whereas 88% of the keeping rows re-quoted their original sentence. The

shortlist does not make nodes invent evidence; it changes the task from "read the passage and answer" to "choose between B and D, then find a sentence that supports the choice", and a passage long enough to answer the question usually contains such a sentence.

Without the passage, a false premise does produce fabrication. When the passage is withheld and the note is a false premise *about the passage*, nodes that follow it sometimes quote sentences that do not exist: 20% (FLASH-A), 22% (FLASH-B) and 65% (PRO) of their responses contain a non-empty quotation that is not in the text, against 0–2% (five nodes) and 8–12% (PRO) under the shortlist notes. The premise form is safer when the node has something to check it against; a confident false claim about evidence the node cannot see is both believed and back-filled.

9 Related work

Misleading context and sycophancy. Several benchmarks show that LLMs change correct answers when the prompt supplies a wrong one: all-wrong option sets (Góral et al., 2025), misleading in-context statements, whether false (Wang et al., 2026) or true but pointing the wrong way (Shafiei et al., 2026), user challenges (Wang et al., 2023), and persuasive misinformation (Xu et al., 2024a). Sycophancy toward user-stated views is documented by Sharma et al. (2024), with training-based mitigations by Wei et al. (2023). Knowledge-conflict studies ask when a model follows context that contradicts its parametric memory (Xie et al., 2024; Xu et al., 2024b), and irrelevant context alone degrades reasoning (Shi et al., 2023). These studies measure the false-note side of our design alone, without a true note on the same item, so they cannot say what a form costs a sender who is sometimes right.

Hints and advice taking. Hints improve LLM accuracy and the effect depends on hint form (Marioriyad et al., 2025); the human advice-taking literature finds that people weight advice at roughly .39 of their own judgement (Bailey et al., 2023). Our rescue rate B is the LLM analogue for wrong advisees.

Abstention and selective prediction. Selective question answering lets a model decline to answer when uncertain (Kamath et al., 2020); the exit clause is the broadcast-side counterpart, letting a receiver with strong reasons decline the note.

Multi-agent debate. Agents that see each other's answers and revise can improve factuality (Du et al., 2024); the same channel propagates manipulated knowledge through a community (Ju et al., 2024), and re-answering without new information does not by itself correct reasoning (Huang et al., 2024), which is why our blank condition keeps $K \approx 1$. Our measures apply to any message written once and read by all; v^*, a break-even probability in the sense of decision analysis (Howard, 1966), bounds how confident the writer should be before posting.

10 Discussion

What a coordinator should post. The answer depends on what the coordinator can say and on how it weighs the two errors. A bare assertion of one answer is the strongest note in both directions: true, it rescues almost every wrong node; false, it destroys almost every right one. A premise moves far fewer nodes either way, and the two break-even probabilities are not distinguishable. When losing a correct answer is worse than missing a rescue, the premise is the form to use: with the echoed protocol a false premise costs a fifth to a quarter of what a false shortlist or a false assertion costs (about three fifths of a false shortlist without the echo), and the true premise rescues about as well as a true shortlist-with-exit. When the coordinator can only name candidates, it should name one if it values rescue as much as keep, and two with an exit clause if it does not; it should never widen the list.

Why the premise is safer. A shortlist asserts the answer's location and nothing else; a node can comply only by moving into the list. A premise asserts a reason; a node can check it against what it knows, and when the reason is wrong the node often keeps its answer. The data are consistent with checking but do not establish it; a weaker push would look the same (§7). The RACE results show that even with the evidence in hand the shortlist is obeyed first and justified second. A node told the answer outright complies almost always, true or false; a node given a reason moves much less often, and whether it moves more for true reasons than for false ones varies by node (Appendix C).

11 Limitations

Scope of items. Our items are English multiple-choice and short open-ended questions from academic subjects and one reading-comprehension corpus; we do not claim the results for code, planning or long-form generation, where an earlier exploratory round found much higher keep rates. The false notes point at one designated wrong option; adversarially optimised notes could do more damage.

Strength. A false shortlist excludes two options with certainty, whereas a false premise is a model-written sentence whose persuasiveness we did not calibrate. The bare-assertion control (§7) does not resolve it: the premise is far safer than the strongest option-format note when false and no cheaper by v^*, a pattern a weaker push would also produce, and the pre-registered test that would have ruled strength out failed. What we still lack is a premise matched to the shortlist in strength, or a length- and specificity-matched pair of premises, so we cannot say how much of the premise's safety comes from its being a reason and how much from its being a weaker push.

Premise quality. Premises were written by one model (DeepSeek V4 Pro), which also serves as a node, and were checked programmatically (no option letters, no gold or target option text, minimum length). After the 297-item round the author hand-audited 40 randomly drawn items and a further 33 flagged because at least two of three strong nodes gave the same wrong first answer (the flag uses only note-free answers). In the random 40, 36 true premises were true and did not state the answer, and 25 false premises were false and read as plausible wrong reasoning, the other 15 being implausible or vacuous; whether a false premise was plausible made no difference to how often it was followed (node-mean keep 83% vs. 84%; shortlist 42% vs. 38%), so the premise's safety is not an artefact of unconvincing false premises. Across both audits ten items were defective: five with a wrong, contested or corrupted answer key or a truncated question, and five with a premise pair that failed in ways the program check does not catch (a premise that states an option, a true premise that points at the target, a false premise that leads to a non-target option). A pre-registered sensitivity analysis drops these ten and recomputes every test of §5–§7 on the remaining 287 items: every direction and decision is unchanged, and no node-mean difference moves by more than .02 (e.g. $v^*(\mathrm{L3}) - v^*(\textsc{cond2})$ on the 297-item set is $-.17$ [$-.29$, $-.01$] before and $-.16$ [$-.28$, $-.00$] after). The random audit was not pre-registered.

Echo. The main rounds echo the node's earlier answer, confidence and reason in the second prompt, so blank keep is $\approx 100\%$ and every effect is measured against a self-anchored prior. The no-echo control (§7) shows that the orderings survive without it, but the echo lowers rescue on every form, lowers keep under a false shortlist and raises it under a false premise, so the premise's keep margin halves without it; the absolute rates in our tables are specific to the echoed protocol.

Statistics. Six nodes per round is a small sample of models, and five of the six are deterministic (temperature 0) with one run per item, so within-node sampling variance is not captured by the item bootstrap; FLASH-B samples at temperature 1 and was also run once. Rescue denominators are as small as 11 (CODEX, 80 items) and 22 (CODEX, 297 items). Node means of a clipped ratio are fragile: one node at exactly 0 can decide the ordering, which is why §6 reports both definitions. Per-node numbers are not stable traits: an attempt to predict which nodes a form would rescue from two fitted coordinates failed to transfer (Appendix B). We report every pre-registered test without correcting for multiple comparisons. Two of the eight nodes in the largest round were lost to quota exhaustion and are excluded by a pre-registered coverage check; the pre-registered minimum of five nodes was met. The control round reuses the 297 items and the same six nodes, so it replicates the protocol but not the item sample.

Judging. Open-ended answers were graded by a model, agreeing with a second model on 92% of a 100-row sample ($\kappa = .83$); it judges the stated conclusion, not the quality of the reasoning.

Ethics statement

All items are from public benchmarks. The false notes were generated for measurement and are released only together with their true counterparts and the gold labels. The findings describe how to write notes that help a team of models; they could also inform someone trying to mislead a model, but the misleading forms we identify (a bare shortlist or a bare assertion) are already the default in

many pipelines and the contribution is to show how to make it safer.

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A Prompts

First answer (multiple choice). "Answer each question." followed by the items, then: `Reply with ONLY a JSON object mapping each question ID to {"answer": "A/B/C/D", "confidence": 0-100, "reason": "one short sentence"}. No prose outside the JSON.`

Second answer. "Here are questions you answered earlier. Give your answer again." followed, per item, by the item text, `Your earlier answer: X (confidence N). Your earlier reason: . . .` and, unless blank, `A note shared with the whole team: . . .`; then the same JSON instruction. In the no-echo variant of the control round the `Your earlier answer` and `Your earlier reason` lines are omitted and nothing else changes.

Open-ended. The item without options; `Here is a task you answered earlier. Give your answer again.` with the earlier free-text answer echoed; the note names option texts ("the correct answer is one of: 36%; 81%."). Judge prompt: `Grade each response against the reference. Mark correct = 1 only if the response's stated conclusion agrees with the reference; hedging without the right conclusion, or a contradictory answer, is 0. Ignore style and extra detail.`

Premise generation. Given the question, options, gold option and target wrong option, DeepSeek V4 Pro was asked for a JSON object with `L3t`: "the true key premise or step from which a careful reader could work out the answer. Do not state the answer or repeat the correct option text." and `L3f`: a matched misleading premise that leads to the target option, under the same constraints.

B A negative result on node-level parametrisation (exploratory)

In earlier rounds we tried to summarise each node by two coordinates fitted on calibration items ("takes the form" and "distinguishes true from false") and to predict which nodes a form would rescue. The coordinates predicted the ordering of nodes on held-out hard items but not the absolute rates, and they did not transfer to open-ended tasks or to non-LLM rule-based nodes. We report this to discourage treating the per-node numbers in our tables as stable traits.

C Per-node results, 297-item round

Node	right	wrong	SET2$_t$	SET2$_f$	COND2$_t$	COND2$_f$	L3$_t$	L3$_f$	L3C$_t$	L3C$_f$
FLASH-A	239	58	100/41	42/5	100/28	94/14	100/33	92/14	100/38	92/16
FLASH-B	240	57	100/44	39/9	99/33	83/12	100/37	84/12	100/40	83/18
PRO	240	57	100/30	5/0	100/30	38/0	100/21	82/0	100/28	72/0
CODEX	275	22	99/27	67/0	100/14	96/0	100/41	99/9	100/50	99/0
QWEN	143	154	86/42	41/2	90/38	44/1	97/22	55/8	99/31	44/8
LLAMA	85	212	96/6	76/0	100/0	99/0	99/3	95/0	100/3	98/0

Table 5: Per node: number of first-right and first-wrong items, and K/B (%) per form. LLAMA is almost never rescued by any note; PRO keeps only 5% under a false shortlist and 38% with the exit clause, yet 82% under a false premise.

	with echo		without echo							
Node	ONE1$_t$	ONE1$_f$	BLANK	SET2$_t$	SET2$_f$	COND2$_t$	COND2$_f$	L3$_t$	L3$_f$	v^*(ONE1)
FLASH-A	100/95	27/2	97/9	98/40	72/3	98/38	84/5	100/55	79/7	.79
FLASH-B	100/98	13/5	92/26	95/53	71/7	97/41	80/9	98/53	74/18	.82
PRO	100/100	0/0	100/7	98/47	38/4	98/49	60/0	99/53	71/14	.81
CODEX	100/64	93/0	100/9	100/36	85/0	100/23	97/5	100/55	99/14	.56
QWEN	99/96	13/3	95/4	90/41	43/3	87/35	62/1	96/38	62/6	.45
LLAMA	100/82	13/0	82/9	67/39	12/3	73/39	29/2	85/37	51/8	.30

Table 6: Control round (§7), same items and nodes: K/B (%) for the bare single-option assertion with echo, and for the main forms with the echo removed; last column is the clipped v^* of the bare assertion. The true assertion rescues almost every wrong node; the false one destroys almost every right one except on CODEX.